TECHNICAL INSIGHTS REPORT

Consumer Behavior Analysis — The Generational Divide in Shopping

Exploratory Storytelling With Data | AltSchool Africa | Data Science Track

Author	**Gabriel Bello-ososo**
Dataset	Consumer Behavior Dataset with 10,500 records
Source	UCI / Kaggle

1. Dataset Characteristics

The dataset is a clean, structured record of consumer purchasing behavior containing 10,500 rows with zero missing values across all ten variables. It covers six generational age cohorts, four product categories of interest, and tracks both transactional and behavioral metrics per customer.

1.1 Overview Statistics

Attribute	Value
Total Records	10,500 customer transactions
Age Groups Analysed	6 cohorts: 18–25, 26–35, 36–45, 46–55, 56–65, 65+
Missing Values	0 — Dataset is fully clean, no imputation required
Subscription Rate	29.3% (3,076 subscribers vs 7,424 non-subscribers)
Gender Split	Female: 5,530 (52.7%) | Male: 4,970 (47.3%)
Product Categories	6: Books, Sports, Beauty, Home & Garden, Electronics, Clothing
Payment Methods	5: Mobile Payment, Debit Card, PayPal, Credit Card, Cash (evenly distributed)
Seasons Covered	Spring, Summer, Fall, Winter

1.2 Age Group Distribution

Age Group	Count	% of Total	Generation Label
18–25	1,930	18.4%	Gen Z
26–35	2,659	25.3%	Millennials (Older)
36–45	2,341	22.3%	Gen X
46–55	1,505	14.3%	Late Gen X / Boomers
56–65	1,242	11.8%	Baby Boomers
65+	823	7.8%	Silent Generation

> Key observation: The 26–35 cohort is the largest segment (25.3%), followed by 36–45 (22.3%). Together, these two groups represent nearly half (47.6%) of all customers in the dataset. The 65+ cohort is the smallest segment, comprising just 7.8% of the customer base.

2. Spending Patterns

Spending analysis examines the purchase amount variable across the full dataset and within each demographic segment to identify where purchasing power is concentrated and what drives variation in transaction value.

2.1 Overall Spending Summary

Metric	Value	Interpretation
Mean Purchase	**$86.01**	Average per transaction
Median Purchase	$67.84	50th percentile — typical buyer
Std Deviation	$64.02	High spread — diverse price range
Minimum	$25.00	Floor price in the dataset
Maximum	$500.00	Top-tier transactions

> The mean ($86) exceeds the median ($67.84), indicating a right-skewed distribution with a long tail of high-value transactions pulling the average upward. The high standard deviation ($64.02) confirms wide spread across purchase amounts — not a uniform spending population.

2.2 Average Spending by Age Group

This is the core analytical finding of the project: spending follows an inverted-U curve across life stages, peaking in the mid-career years and declining toward retirement.

Age Group	Avg Purchase	vs. Overall Avg	Interpretation
18–25	$61.43	−25%	Entry-level income; student budgets
26–35	**$88.77**	**+8%**	Career building; income rising
36–45 ★	**$110.94**	**+36%**	**PEAK — established careers, family spending**
46–55	**$100.35**	**+23%**	Strong — near-peak earning years
56–65	$75.90	−7%	Transitioning; more budget-conscious
65+	$52.80	−35%	Lowest — fixed income, deliberate spending

> The 36–45 age group spends 110.1% more per transaction than the 65+ group — more than double.
> Three cohorts exceed the overall average ($81.70): 26–35, 36–45, and 46–55.
> This inverted-U pattern mirrors the classic life-cycle income hypothesis in consumer economics.

3. Purchase Frequency Patterns

Purchase frequency (annual purchases) captures how often each customer shops per year. Unlike spending, frequency tells a markedly different generational story: younger consumers shop far more often, not more per transaction. This reveals an inverse relationship between age and engagement level.

3.1 Overall Frequency Statistics

Mean Annual Purchases	6.2 purchases per year
Highest Recorded (individual)	14 purchases per year
Lowest Recorded (individual)	1 purchase per year

3.2 Frequency by Age Group

Age Group	Avg Annual Purchases	vs. 65+ Baseline	Shopping Behaviour Profile
18–25	**7.0x / yr**	**+181%**	Digital-native; habitual online shoppers
26–35 ★	**9.0x / yr**	**+261%**	**HIGHEST frequency — subscription & flash-sale driven**
36–45	**6.0x / yr**	**+141%**	Balanced — intentional but regular
46–55	4.5x / yr	+81%	Moderate — planned purchasing
56–65	3.5x / yr	+41%	Low frequency; deliberate over impulse
65+	2.5x / yr	Baseline	Lowest — careful, infrequent transactions

> The 26–35 age group shops 3.62× more frequently than the 65+ cohort (9.0x vs 2.5x per year).
> Note the paradox: the highest-frequency group (26–35) is NOT the highest spender per transaction (36–45).
> This split between frequency and transaction value demands different strategies for each segment.

3.3 The Frequency–Spend Paradox

A critical insight emerges when frequency and spending are examined together: the groups with the highest transaction frequency (18–35) are NOT the highest per-transaction spenders. Conversely, the group with the highest per-transaction value (36–45) is only a moderate-frequency shopper.

Segment	Avg Spend / Transaction	Avg Annual Frequency	Strategic Focus
18–35 (Young Adults)	$61–$89 (lower)	**7–9x / yr (highest)**	Frequency — subscriptions, flash sales
36–45 (Prime Years)	**$111 (peak)**	6x / yr (moderate)	Value — premium products, quality
56+ (Mature Adults)	$53–$76 (lower)	2.5–3.5x / yr (lowest)	Curation — quality, clarity, trust

4. Seasonal Purchase Analysis

Seasonal analysis examines total purchase volume by season to identify whether specific times of year drive meaningfully higher consumer spending. The findings reveal a surprisingly flat pattern — a useful signal in itself.

4.1 Total Purchase Volume by Season

Season	Total Purchase ($)	% of Annual Total	Relative Rank
Spring ★	**$231,153.80**	**25.8%**	**1st — Highest**
Winter	$226,709.18	25.3%	2nd
Fall	$226,223.18	25.2%	3rd
Summer	$218,967.48	24.4%	4th — Lowest
TOTAL	**$903,053.64**	100%	—

> The range between highest (Spring: $231,154) and lowest (Summer: $218,967) season is only $12,186 — a spread of just 5.6%.
> This flat distribution indicates consumer demand is year-round and not strongly seasonal.
> Season alone is an unreliable targeting lever; it should be combined with age-group data for meaningful segmentation.

5. Subscription Impact Analysis

29.3% of customers hold active subscriptions (3,076 out of 10,500). The subscription cohort is a strategically important segment: subscribers demonstrate both higher spending per transaction and higher annual purchase frequency, creating a compounding effect on total lifetime value.

5.1 Subscriber vs. Non-Subscriber Comparison

Metric	Subscribers (29.3%)	Non-Subscribers (70.7%)	Uplift
Customer Count	3,076	7,424	—
Avg Purchase ($)	**$90.93**	$83.96	**+8.3%**
Avg Annual Purchases	**6.9x / yr**	5.8x / yr	**+19.1%**

> Subscribers spend $6.97 more per transaction (+8.3%) AND shop 1.1 times more per year (+19.1%).
> The compound effect: a subscriber generates approximately $627 / yr vs a non-subscriber's $487 / yr in annual revenue
> — a 28.7% lifetime value advantage, simply from holding a subscription.

5.2 Subscription Rate by Age Group

Subscription adoption is not uniform across generations. The 26–35 cohort shows the highest subscription rate, aligning with their digital-native behavior and comfort with recurring payment models.

Age Group	Subscription Rate	Relative Rank	Insight
26–35 ★	44.7%	Highest	Prime subscription target demographic
36–45	37.7%	2nd	Strong adoption — high-value sub members
18–25	23.8%	3rd	Budget-constrained; growth opportunity
46–55	22.5%	4th	Below avg adoption; consider targeted trials
56–65	11.8%	5th	Low adoption; prefer one-time purchases
65+	7.3%	Lowest	Minimal adoption — trust & simplicity needed

6. Product Category Analysis

Six product categories are represented in the dataset. Average spend per category is relatively narrow, with a range of only $4.21 between highest and lowest. Therefore, suggesting that category type alone is a weak differentiator for pricing strategy. The strategic value lies in the category preferences by age group.

6.1 Average Spend by Category

Category	Avg Purchase ($)	Rank	Notable Characteristic
Books ★	$88.10	1st	Highest avg spend; educational/perceived-value purchases
Sports	$86.81	2nd	Active lifestyle correlation with higher willingness to pay
Beauty	$86.43	3rd	Strong repeat-purchase category; brand loyalty driven
Home & Garden	$85.43	4th	Seasonal peaks expected in Spring and Fall
Electronics	$85.32	5th	High-intent, considered purchases; lower purchase frequency
Clothing	$83.89	6th	Highest transaction volume, lowest avg spend

> Category spread is narrow: only $4.21 separates Books ($88.10) from Clothing ($83.89).
> Category-level pricing strategy has limited leverage — demographic targeting is more impactful than category focus alone.

6.2 Top Category by Age Group

Cross-referencing product category with age group reveals distinct generational preferences, providing a foundation for personalized product recommendation engines.

Age Group	Top Category	Avg Spend in Category	Implication
18–25	**Books**	$62.99	Educational purchases; digital-first content
26–35	**Books**	$93.38	Career development spend; highest book spend
36–45	**Beauty**	**$113.44**	Premium beauty; self-care investment
46–55	**Sports**	$105.96	Health-conscious spending; active lifestyle
56–65	Electronics	$81.67	Utility-driven; assistive technology
65+	Electronics	$55.21	Practical, functional purchases

7. Customer Review Ratings

Review ratings provide a proxy for overall customer satisfaction. The dataset contains three rating levels which are: 3, 4, and 5. And the distribution is skewed toward positive scores.

Rating	Count	% of Reviews	Interpretation
5 — Excellent	4,220	**40.2%**	Largest positive group
4 — Good	4,740	**45.1%**	Strong satisfaction
3 — Average	1,540	14.7%	Improvement opportunity
Mean Rating	**4.26 / 5.0**	—	Strong overall sentiment

> 85.3% of customers rated their experience 4 or above — indicating strong overall satisfaction.
> Only 14.7% of reviews are at the 3-star level; no 1 or 2-star ratings exist in this dataset.
> High satisfaction scores support the viability of subscription upselling and repeat-purchase strategies.

8. Key Technical Findings Summary

The following findings are drawn directly from quantitative analysis of the dataset and are the analytical foundation for the strategic recommendations presented in the accompanying presentation.

Finding 1 — Spending Peaks at 36–45, Not at Peak Frequency

- The 36–45 age group spends 110.1% more per transaction than the 65+ group.
- The inverted-U spending curve mirrors the classical economic life-cycle hypothesis.
- Three cohorts exceed the $81.70 overall average: 26–35, 36–45, and 46–55.

Finding 2 — Young Shoppers Lead in Frequency, Not Value

- 26–35 year-olds purchase 9x per year: 3.62× more often than the 65+ cohort.
- Frequency and per-transaction value are inversely related across generations.
- This bifurcation demands entirely different marketing strategies for each segment.

Finding 3 — Seasonality Is Weak; Demand Is Year-Round

- The top season (Spring: $231,154) exceeds the lowest (Summer: $218,967) by only 5.6%.
- All four seasons contribute approximately 25% of annual purchase volume.
- Season should be combined with demographic data, not used as a standalone targeting signal.

Finding 4 — Subscribers Are Disproportionately Valuable

- Subscribers spend 8.3% more per transaction AND purchase 19.1% more frequently.
- This compounds to approximately a 28.7% annual revenue advantage per subscriber.
- 26–35 cohort has the highest subscription rate (44.7%), so they are the prime conversion target.
- 65+ subscribers have only 7.3% adoption, indicating that significant trust and simplicity barriers exist.

Finding 5 — Category Spend Is Narrow; Demographic Targeting Dominates

- Only $4.21 separates the highest-spend category (Books: $88.10) from the lowest (Clothing: $83.89).
- Category preferences vary distinctly by age: 18–35 prefer Books, 36–45 prefer Beauty, 56+ prefer Electronics.
- Personalized recommendations by demographic will outperform category-level pricing strategies.

9. Methodology & Technical Notes

9.1 Tools & Libraries

- Python 3 — primary analysis language
- Pandas — data loading, groupby aggregations, cross-tabulations
- NumPy — statistical computations (Pearson correlation, polynomial fitting)
- Matplotlib & Seaborn — visualization library for all charts

9.2 Analysis Approach

All analyses were performed on the cleaned dataset (no rows removed, no imputation needed). Descriptive statistics were computed using Pandas groupby on the primary categorical variables (Age Group, Season, Category, Has Subscription). Correlation analysis between Annual Purchases and Purchase Amount was computed using the Pearson method split by subscription status.

9.3 Data Quality Statement

- Total records: 10,500 | Missing values: 0 | Duplicate rows: not detected
- All categorical variables have consistent, clean labelling — no encoding corrections required.
- Payment Method is approximately uniformly distributed (~2,000 per method) — no payment bias.
- Review ratings only contain values 3, 4, and 5 — no extreme negative scores in the dataset.

Consumer behavior isn't uniform — it's generational.

The businesses that win will understand, target, convert, and optimize.